PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	RECIST 1.1 versus mRECIST for assessment of tumour response to molecular targeted therapies and disease outcomes in patients with hepatocellular carcinoma: a systematic review and meta-analysis
AUTHORS	Yu, Hongli; Bai, Yuping; Xie, Xiaoyu; Feng, Yuemin; Yang, Yao; Zhu, Qiang

VERSION 1 – REVIEW

REVIEWER	Tovoli, Francesco University of Bologna
REVIEW RETURNED	28-Jun-2021

GENERAL COMMENTS	I read with pleasure this paper about the prognostic abilities of RECIST 1.1 and mRECIST in patients with hepatocellular carcinoma (HCCC) treated with anti-angiogenic drugs. The topic is not entirely novel, but remains of interest to the readers. The Authors demonstrate that the rate of progressors is the same regardless of the adopted criteria, but the mRECIST identify a larger subgroup of partial responders qith an higher survival. Overall, the analyses have been performed correctly. The conclusions are largely justified by the results, but there are some points deserving attention: 1) Discussion. From a clinical perspective, the most important quality required by these criteria is to discriminate progressors vs nonprogressors. This distinction allow clinicians to switch to a second-line treatment in case of progression. Discriminating stable disease from response is informative in terms of prognosis but does not change the management of the patient (who will continue the current drug). The Authors should be appraised for distinguishing these aspects in their results, showing a perfect concordance for progression and imperfect concordance for response. However, these aspects should be more stressed in the Discussion, too. 2) Results. Three-group comparisons (progressive disease vs stable disease vs response) both for RECIST and mRECIST would stenghten the Authors' findings and are welcome. 3) Discussion. A common criticism toward mRECIST in similar papers is that they are more difficult to evaluate (due to the need for vascularisation assessment) and that they are difficult to use in clinical practice. The Authors could discuss this point, also in the light of evidence showing that this criticism is actually not true (DOI: 10.1007/s00330-018-5393-3)

REVIEWER	Sun, Ryan The University of Texas MD Anderson Cancer Center Department of Biostatistics
REVIEW RETURNED	21-Aug-2021

GENERAL COMMENTS	The authors present an interesting study comparing the use of RECIST 1.1 and mRECIST for assessing tumor response in HCC. They find that when comparing OS between responders and non-responders according to mRECIST, there is a significant difference in survival. However there is no significant difference when comparing responder and non-responders according to RECIST 1.1. I find the description of the statistical analysis to be a little short and vague. In particular, it is still not clear to me how the HRs were meta-analyzed - were they simply taken from the study text? If so, was the reporting of the HR between responders and non-responders one of the inclusion criteria? There is also a lack of attempt to address possible major confounders, for instance the treatment given or the date of the study. Some phrases are difficult to interpret, e.g. "Our results showed the results were reliable..." Additionally there are scattered typos throughout the text.

REVIEWER	Zhong, Jian-Hong Affiliated Tumor Hospital of Guangxi Medical University
REVIEW RETURNED	11-Dec-2021

GENERAL COMMENTS	1. 'EMBASE, PubMed, Web of Science and Cochrane Library were used to search the studies from inception to 30 June 2020.' Searching need to be update. 2. Many important RCTs of molecular targeted therapies in HCC were not included. 3. All their findings were not novel.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Dr. Francesco Tovoli, University of Bologna
Comments to the Author:
I read with pleasure this paper about the prognostic abilities of RECIST 1.1 and mRECIST in patients with hepatocellular carcinoma (HCCC) treated with anti-angiogenic drugs. The topic is not entirely novel, but remains of interest to the readers. The Authors demonstrate that the rate of progressors is the same regardless of the adopted criteria, but the mRECIST identify a larger subgroup of partial responders qith an higher survival.
Overall, the analyses have been performed correctly. The conclusions are largely justified by the results, but there are some points deserving attention:
Thank you for your positive comments.
1) Discussion. From a clinical perspective, the most important quality required by these criteria is to discriminate progressors vs nonprogressors. This distinction allow clinicians to switch to a second-line treatment in case of progression. Discriminating stable disease from response is informative in terms of prognosis but does not change the management of the patient (who will continue the current drug).

The Authors should beappraised for distinguishing these aspects in their results, showing a perfect concordance for progression and imperfect concordance for response. However, these aspects should be more stressed in the Discussion, too.
Thanks for the suggestion. We have expanded on the original in the fourth paragraph of the discussion section. We discussed the same ability of mRECIST and RECIST1.1 to assess disease progression, which then leads to their synchronicity in guiding clinicians to change second-line drugs.
2) Results. Three-group comparisons (progressive disease vs stable disease vs response) both for RECIST and mRECIST would stenghten the Authors' findings and are welcome.
In the agreement analysis that included data from eight studies, the kappa value for DCR was as high as 0.9. Normally we consider a kappa value greater than 0.8 to mean that there is no difference between the two criteria, which also means that the DCR evaluated by the two criteria are two almost identical groups, so no further analysis of the survival data was carried out.
3) Discussion. A common criticism toward mRECIST in similar papers is that they are more difficult to evaluate (due to the need for vascularisation assessment) and that they are difficult to use in clinical practice. The Authors could discuss this point, also in the light of evidence showing that this criticism is actually not true (DOI: 10.1007/s00330-018-5393-3)
Thanks for your advice. We have added this part after carefully reading the article you mentioned in fourth paragraph of discussion.

Reviewer: 2
Dr. Ryan Sun, The University of Texas MD Anderson Cancer Center Department of Biostatistics
Comments to the Author:
The authors present an interesting study comparing the use of RECIST 1.1 and mRECIST for assessing tumor response in HCC. They find that when comparing OS between responders and non-responders according to mRECIST, there is a significant difference in survival. However there is no significant difference when comparing responder and non-responders according to RECIST 1.1.
We thank for the positive comments.
I find the description of the statistical analysis to be a little short and vague. In particular, it is still not clear to me how the HRs were meta-analyzed - were they simply taken from the study text? If so, was the reporting of the HR between responders and non-responders one of the inclusion criteria? There is also a lack of attempt to address possible major confounders, for instance the treatment given or the date of the study.
Thanks for pointing it out. We have made certain additions to the statistical analysis section. The HRs were extracted from the text or from the K-M curves by the software Engauge Digitizer. We therefore summarized the HRs obtained by direct and indirect means in the inclusion criteria by saying "available data on overall survival and K-factor or sufficient information to calculate it".
Some phrases are difficult to interpret, e.g. "Our results showed the results were reliable.." Additionally there are scattered typos throughout the text.
Thanks for the suggestion. We have performed the English language editing on this manuscript.

Reviewer: 3
Dr. Jian-Hong Zhong, Affiliated Tumor Hospital of Guangxi Medical University
Comments to the Author:
1. 'EMBASE, PubMed, Web of Science and Cochrane Library were used to search the studies from inception to 30 June 2020.' Searching need to be update.
Thank you for your advice. We have updated our literature search. A comprehensive search of PubMed, EMBASE, Web of Science, and the Cochrane Library from inception through December 31, 2021 was performed.
2. Many important RCTs of molecular targeted therapies in HCC were not included.
During our search we noted a number of important RCTs of molecularly targeted therapies for HCC, which were also discussed. However, they were limited by the fact that some of these studies have not published complete findings or have not used both mRECIST and RECIST1.1 to evaluate post-

treatment tumor response. Nevertheless, some of the RCTs met the inclusion criteria, but only qualitative analyses were performed because the required data were missing.
3. All their findings were not novel.
Molecularly targeted agents are a new approach to the treatment of hepatocellular carcinoma that has received widespread attention in recent years and is increasingly being used in clinical practice. Current methods for assessing systemic therapy are mainly mRECIST and RECIST1.1, but there is still a lack of consensus on the efficacy of both assessment criteria in evaluating tumour response after treatment with molecularly targeted agents. We included 2,563 patients from a total of 23 studies published between 2021 and 2021 in the qualitative analysis and 566 patients from seven studies in the meta-analysis.
We found that mRECIST appears to be more accurate than RECIST1.1 in assessing objective response rate after molecular targeted therapies in HCC patients and can better assess the prognosis between patients in the objective response group and those in non-objective response. However, the performance of both criteria in assessing disease progression is approximately equal. We believe that our study has practical implications for clinical practice, particularly in the evaluation of whether to switch to second-line drugs.

VERSION 2 – REVIEW

REVIEWER	Tovoli, Francesco University of Bologna
REVIEW RETURNED	24-Mar-2022

GENERAL COMMENTS	The Authors addressed all of my points. I have no further comments.